

Data Exchange Protocol
for
Andesfit 4G Blood Pressure Monitor

V2.0.0

Andesfit System Limited

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AMENDMENT HISTORY

Version No.	Summary of Changes	Date	Drafted
2.0.0	Added Reminder, X-API-Key feature, updated blood pressure json structure(added CSQ and BAT)	2 Apr 2025	MC
1.2.0		26 Jul 2024	MC
1.1.0	Page 5: Basic workflow diagram Wireless icon P. 6: “during production” P. 8: “SN” key; reupload workflow	16 Jul 2024	MC
1.0.0	Initial Release	3 June 2024	MC

Contents

Introduction

Purpose of document	4
Overview of device	4

Device Description 4

Basic Workflow	5
Taking a Blood Pressure Measurement	5

Communication Protocols

Supported communication protocol	6
URL for API Endpoint	6
JSON Payload	7
Expected Server Response	8

Device Configuration

Setting the Internal Clock	9
Sample Command	10
Setting the Reminder Time	11
Setting the Api Key	12

Troubleshooting 13

Introduction

Purpose of the Document

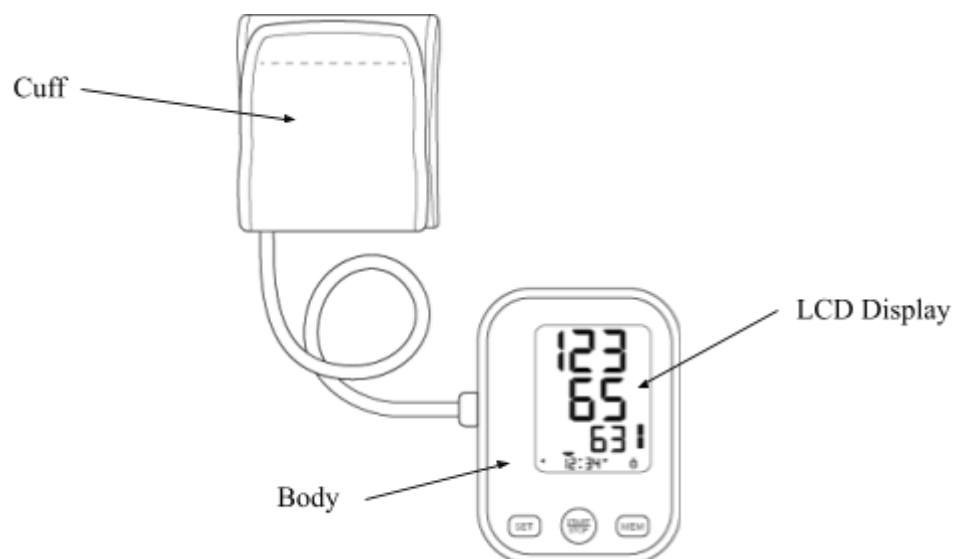
The purpose of this document is to define the communication protocol for the Andesfit 4G Blood Pressure Monitor (BPM). This document serves as a reference for integrating this BPM to other systems with internet connectivity, and provides guidelines for developers and system integrators on decoding.

Overview of Device

The Andesfit 4G Blood Pressure Monitor, Model: ADF-H62, is able to take blood pressure measurement of a user and, upon each measurement, transmit the result to a fixed internet URL via 4th Generation(“4G LTE”) broadband cellular network.

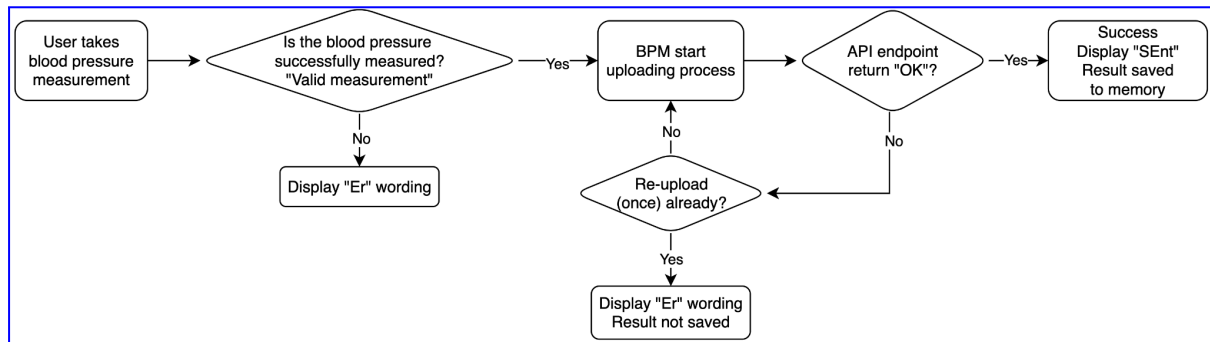
Device Description

Each BPM is equipped with a main body with a cuff connected by a soft rubber air duct. The product is operated by using the buttons on the surface of the body as shown below:



Basic Workflow Diagram

This diagram shows the basic scenario of the measurement-to-upload process:



Taking a Blood Pressure Measurement

Before measurement:

1. Make sure the batteries are properly installed to the BPM
2. Make sure the rubber tubing of the cuff is attached to the BPM
3. Be seated as shown below
4. Put on the cuff onto your LEFT UPPER ARM, 1-2 cm above the elbow crease line, the rubber tubing should be aligned to the center of the elbow crease line, pointing to the direction of the lower arm (See picture below)
5. Tighten the cuff so that it is snug comfortably to the upper arm, and that you can fit 2 fingers between the cuff and the upper arm. Secure the cuff with the velcro
6. Place the arm on the table and let your body relax for 1 minute

Taking a measurement:

1. Tap the “Start/Stop” button once, the display of the BPM should come on, showing the date and time. Speech guidance will start playing and the measurement process will start automatically
2. Remain calm and seated, breathe normally. Avoid moving your body or talking during the measurement
3. If the measurement is valid, the result will be shown on the display at the end of the measurement
4. The “” icon will start blinking and the BPM will start connecting to the API endpoint and upload the result. Wait until the “SEnt”(for successful upload) or “Er”(upload error/failed) wording is shown on the display
5. Tap the “Start/Stop” button once to turn off the BPM



Communication Protocol

The 4G Blood Pressure Monitor (BPM) follows the below protocol when uploading measurement result:

Protocol & Encryption		HTTPS only
Request Type		POST
Headers	Content-type	JSON (application/json)
	Encoding/Charset	UTF-8
	X-API-Key	A 32 Character String (see below)

URL for API Endpoint

The URL for the API endpoint that receives the measurement result is FIXED and unchangeable once assigned to the device [during the manufacturing stage](#). User identification, data forwarding or other procedures should be handled by the API endpoint after receiving the uploaded data.

URL length	128 characters or below
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Examples:

`https://www.example.com/api/health`

`https://123.123.123.123/data-uploads-CaNtBeGuEsSeDByBoTs123XyZ`

Please provide the exact URL format, endpoint name, and any other relevant details that the BPM should use when uploading data to your API server.

X-API-Key

The X-API-Key is a configurable 32 character string in the header for API servers to distinguish if the incoming POST is from an actual device. This key can be configured using a specific server response. See Page 12.

Sample JSON Payload

A typical measurement result post by the BPM is as follows when formatted:

```
{
  "imei": "861234567890123",
  "user": "1",
  "sys": "188",
  "dia": "080",
  "pul": "063",
  "ano": "0",
  "time": "2023-05-31/07:19:22",
  "SN": "2024042200002",
  "iccid": "01234567890123456789",
  "imsi": "460045678901234",
  "CSQ": "17",
  "BAT": "075"
}
```

The following table describes the JSON structure and the meaning of each key:

Key	Type	Description
imei	string	IMEI of the product
user	string	Value is always “1”. Can be ignored.
sys	string	Systolic pressure in mmHg. 3 digits, zero-filled. Example: “188” = 188mmHg
dia	string	Diastolic pressure in mmHg. 3 digits, zero-filled. Example: “080” = 80mmHg
pul	string	Pulse in bpm (beats-per-minute). 3 digits, zero-filled. Example: “063” = 63 bpm
ano	string	Abnormal heartbeat “0” - Normal, “1” - Abnormal
time	string	Date and time of the measurement

		Format: "YYYY-MM-DD/HH:MM:SS" YYYY: Calendar Year in 4 digit format MM: Month in 2 digit format, zero-filled DD: Day in 2 digit format, zero-filled HH: Hour in 24-hour format, zero-filled MM: Minute in 2 digit format, zero-filled SS: Second in 2 digit format, zero-filled "- / ": Separators Example: "2023-05-31/07:19:22" = May 31, 2023 7:19:22 AM
SN	string	Reference Serial Number of the device (note: the key is in capital letters. Should not to be treated as the product serial number)
iccid	string	ICCID of the installed 4G SIM
imsi	string	IMSI, for reference only
CSQ	string	4G Signal Strength value: 0-31(strongest)
BAT	string	Battery level: "005, 025, 050, 075, 100"(100 = full)

Expected Server Response

The BPM expects the server to respond within 1 minute and a.s.a.p upon reception of the measurement result:

Header

Status Code	200
Content-type	Plain Text (text/plain or text/html)
Encoding/Charset	UTF-8

Response Body

```
OK&reminder=14:20:1#end#
```

Effect

BPM shows "Sent" wording on the display after receiving this response. The measurement result is then saved into the BPM's internal memory and the reminder time is set to 14:20. If the status code is not '200' and the response body does not begin with 'OK', the BPM will treat the upload as failed and shall re-upload once again. If the reupload also fails the result will not be saved in the internal memory.

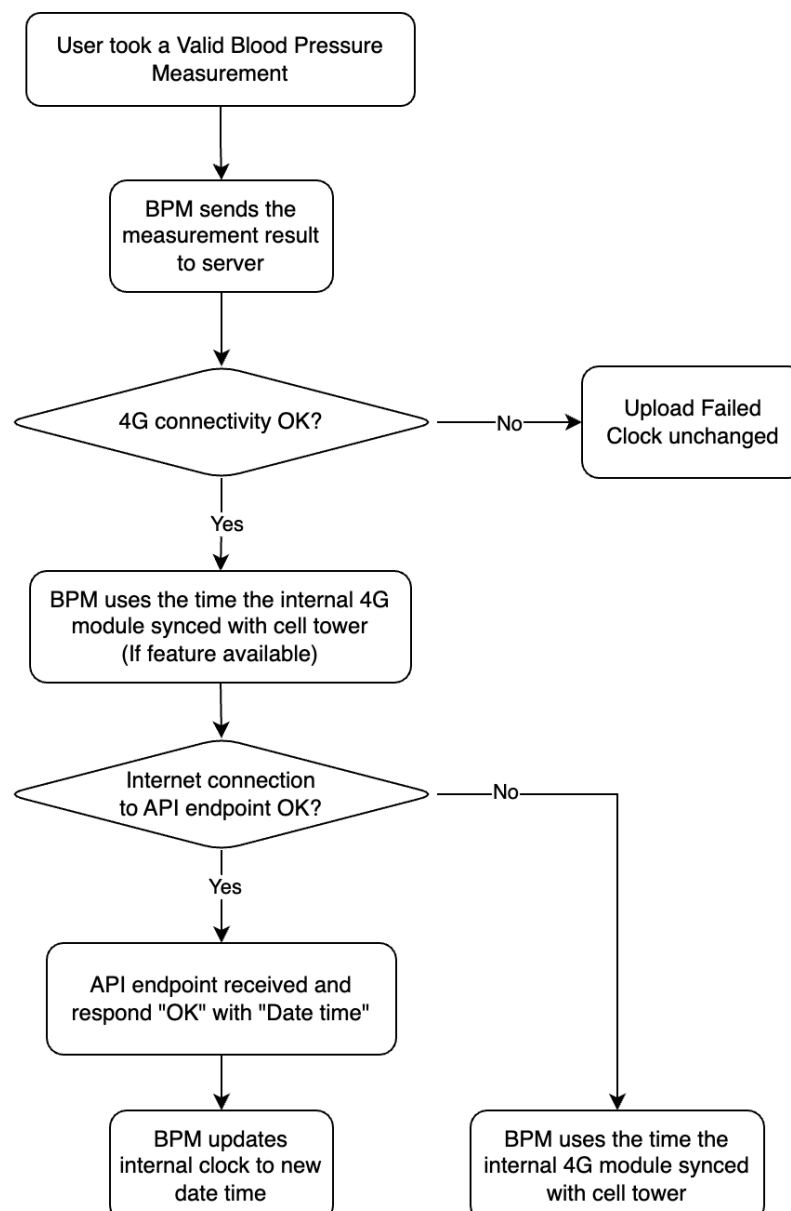
Device Configuration

Setting the Internal Clock

The BPM accepts a clock-setting command by specifying the date and time in a server response for uploads. This is useful to keep the uploaded records' date and time in sync with the API server.

Note: The internal clock resets each time the batteries are replaced, BPM will first use the time synced if the 4G cell tower offers this feature. If the server response has this clock-setting command, the BPM then updates the clock with the date time offered.

Workflow Diagram



Sample Command

The API endpoint received the measurement result from BPM and offer the below response:

Header

Status Code	200
Content-type	Plain Text (text/html)
Encoding/Charset	UTF-8

Response Body

OK202305310719&reminder=14:20:1#end#

Format

OKYYYYMMDDHHMM&reminder=hh:mm:state#end

OK: "OK", means measurement result received

YYYY: Calendar Year in 4 digit format

MM: Month in 2 digit format, zero-filled

DD: Day in 2 digit format, zero-filled

HH: Hour in 24-hour format, zero-filled

MM: Minute in 2 digit format, zero-filled

hh: Reminder Hour in 24-hour format, zero-filled

mm: Reminder Minute in 2 digit format, zero-filled

State: Reminder switch: 0 = off, 1=on

Effect

The BPM will set the clock to May 31, 2023 7:19:00 AM, and the reminder clock is also set to 14:20 after receiving this response and an alarm will trigger by 14:20, after receiving this response.

Setting the Reminder Time

The BPM supports a reminder clock that triggers at a specific time in a day to remind the user to take a measurement. The reminder could be switched ON or OFF by the server response. This part is compulsory following the “OK” text in every response.

Sample Command

Header

Status Code	200
Content-type	Plain Text (text/html)
Encoding/Charset	UTF-8

Response Body

```
OK&reminder=14:20:1#end#
```

Format

```
OK&reminder=hh:mm:state#end
```

OK: “OK”, means measurement result received

hh: Hour in 24-hour format, zero-filled

mm: Minute in 2 digit format, zero-filled

state: 0 = off, 1=on

Effect

The BPM will set the reminder clock to 14:20 after receiving this response and an alarm will trigger by 14:20.

Changing X-API-Key

By supplementing the server response with &Key=xxx, the X-API-Key in the device will be updated. It is recommended to use a table to keep track of previous api keys for a device, since the response might be lost during poor 4G reception or other network problems, causing the device's key to remain unchanged.

Sample Command

The API endpoint received the measurement result from BPM and offer the below response:

Header

Status Code	200
Content-type	Plain Text (text/html)
Encoding/Charset	UTF-8

Response Body

```
OK&key=a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6#end&reminder=14:20:1#end#
```

“a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6” is the default x-api-key value for a new product.

Format

```
OK&key={{32 character alphanumerical key}}&reminder=hh:mm:state#end
```

IMPORTANT: The sequence of the response string should be positioned as shown below.

For example, when the date-time, the reminder time and api-key are all needed to be updated:

Response Body

```
OK202503210820&key=f36f88c14455fe26aa84f7fb5de1d774#end#&reminder=14:20:1#end#
```

Format

```
OKYYYYMMDDHHMM&key={{new api key}}#end&reminder=hh:mm:state#end
```


Troubleshooting

<i>Problem: “Cannot see the measurement result at the API endpoint”</i>	
Symptoms	Possible Reason(s) / Solution
No display on BPM when tapping the “Start/Stop” key, no “air-pump” sound or speech is heard from the BPM	BPM might be out of battery > Check that the battery are installed correctly > Use new batteries & make sure the battery type is correct
Blood pressure is taken but “Er” wording showed up on display during or at the end of the measurement, the result did not show up on the display	BPM could not measure a valid blood pressure > Check that the user is wearing the cuff properly > Check that the user is remaining calm, seated and stationary during the measurement > Measurement upload only take place when a valid measurement is taken
Blood pressure is taken correctly and the result showed up on the display, the signal icon “	